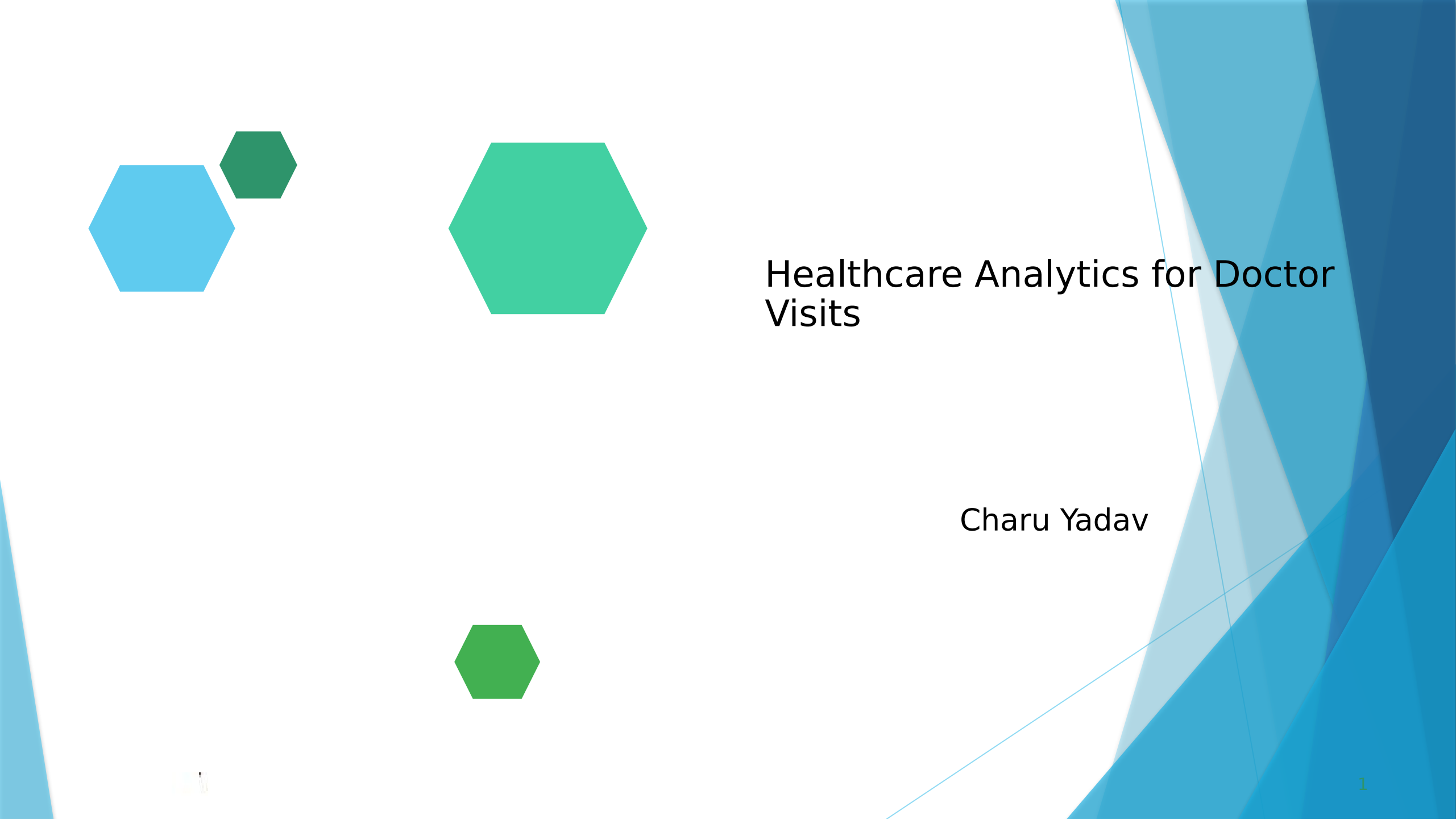


The slide features a white background with several decorative elements. On the left, there are four hexagons: a light blue one, a dark green one, a large medium green one, and a small green one. On the right, there are abstract, overlapping blue geometric shapes. The title 'Healthcare Analytics for Doctor Visits' is positioned in the upper right area.

Healthcare Analytics for Doctor Visits

Charu Yadav

PROBLEM STATEMENT

- Healthcare providers and policymakers often assume that expanding insurance coverage automatically increases access to care. But it's unclear which factors actually drive how often people visit the doctor — is it insurance type, income, chronic health conditions, or demographics?
- This project analyses real health survey data to uncover what truly predicts doctor visits, and tests common assumptions about insurance and access against the evidence.



Project Description



This project analyses 5,190 individuals from an Australian health survey, examining the number of doctor visits alongside demographics, income, health status, insurance type, and chronic condition indicators. Using Python (Pandas, Seaborn, SciPy) in Jupyter Notebook, the data was cleaned, rescaled for interpretability, and explored through univariate, bivariate, and multivariate analysis.

Statistical tests (t-tests, chi-square) were used to confirm which patterns are real versus due to chance, followed by four student-designed deep-dive analyses into insurance effects and visit behaviour.



WHO ARE THE END USERS?



Healthcare policymakers designing insurance and access programs
Insurance providers evaluating who actually uses healthcare services
Public health researchers studying access and utilization patterns
Data analytics students studying real-world healthcare data



Technology Used

Python 3

Pandas & NumPy - data cleaning and analysis

Matplotlib & Seaborn - visualizations

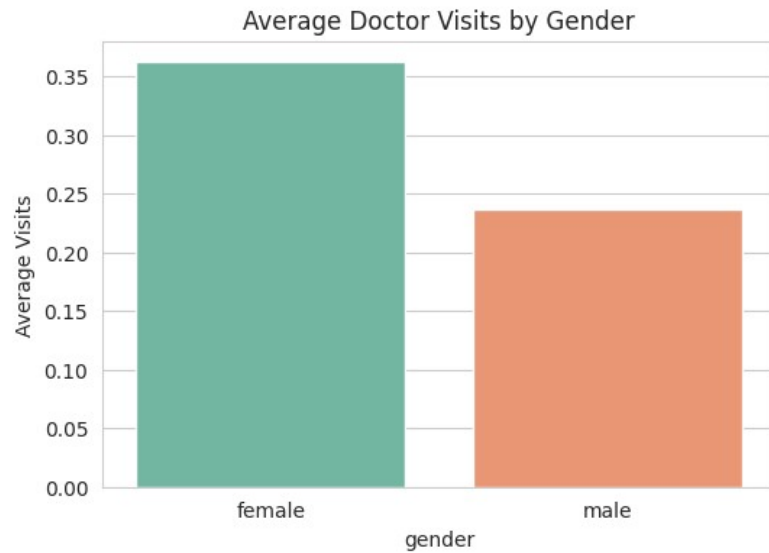
SciPy - statistical testing (t-tests, chi-square)

Jupyter Notebook

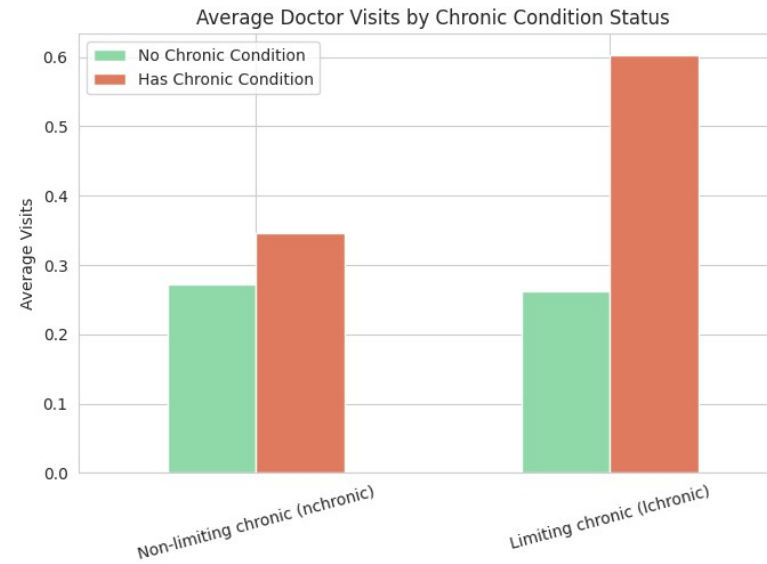


RESULTS

Visits by Gender



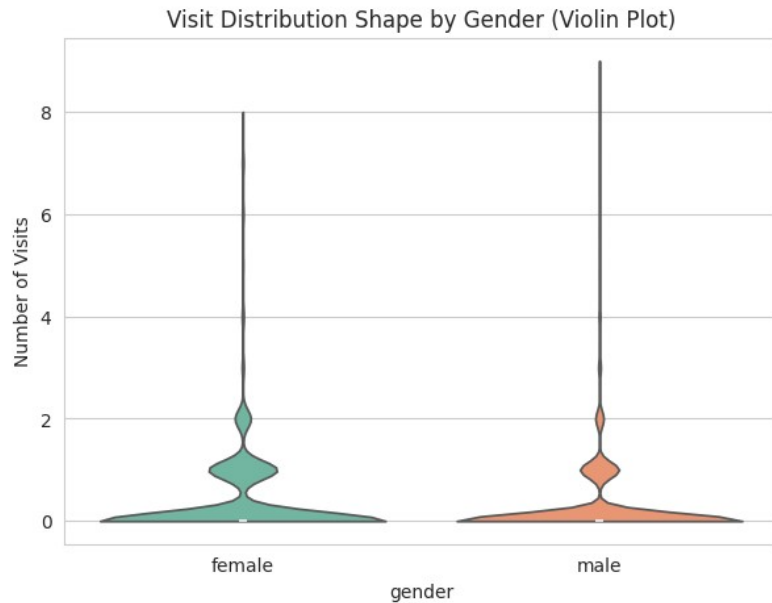
Chronic Condition



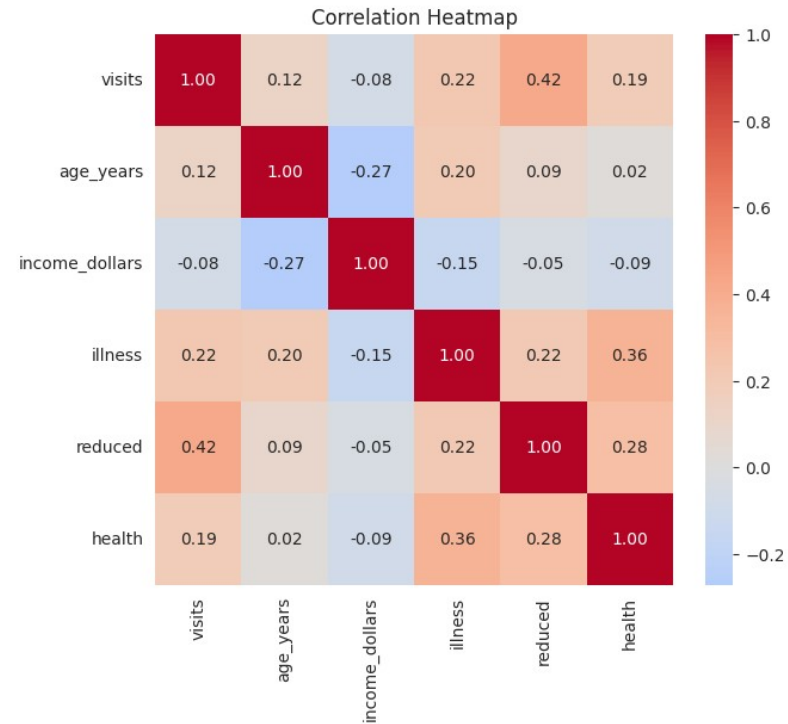
[Demo Link](#)

RESULTS

Visits by Gender (Violin)



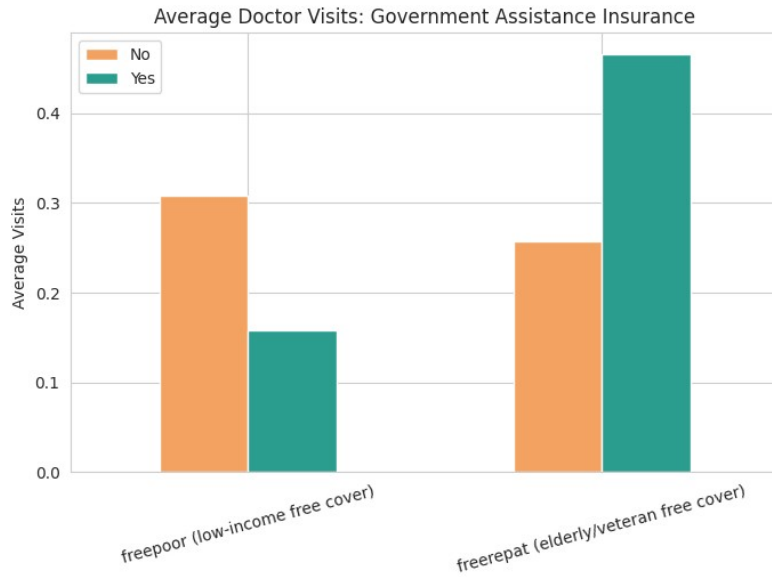
Correlation Heatmap



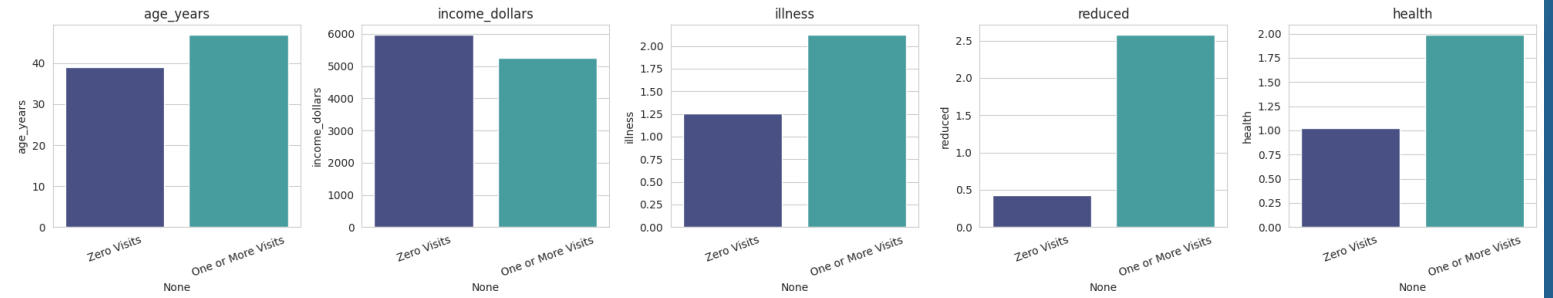
[Demo Link](#)

RESULTS

Government Insurance



Zero-Visit vs Any-Visit



[Demo Link](#)

Thank you

Charu Yadav